RUNGTA College of Engineering and Technology, Bhilai, C.G.

Course: **B. Tech** Sem: **VI (6A)**
Branch: **CSE – AI** Session: **2025 - 26**
Subject: **COMPILER DESIGN**
Subject Code: **C109611(22)**
Max. Marks: **20** Date of Submission: **08/04/2026**

Assignment - 3

Note: -All questions are compulsory and carry 5 marks each.

Sl. No.	Question	CO	Bloom's Taxonomy Level
Q1.	Draw the annotated parse tree for the expression: $(3 + 4) * (5 + 6)$ OR $2 + 3 * 7$	CO3	CR
Q2.	Represent the expression $x = (a + b) * + -c / d$ OR $Q := -b * (c + d)$ using: • Quadruples • Triples • Indirect Triples	CO3	AP
Q3.	Translate the expression $-(a + b) * (c + d) + (a + b + c)$ into Three-Address Code (3AC).	CO3	AP
Q4.	Generate the three-address code for the Boolean expression: a < b OR c < d AND e < f using the backpatching technique.	CO3	CR
Q5.	Construct the syntax tree and postfix notation for the expression: $(a + (b * c) \wedge d - e / (f + g))$	CO3	AN + CR

Bloom's Taxonomy Levels/Phrases: Remembering (RE), Understanding (UN), Applying (AP), Analysing (AN), Evaluating (EV) & Creating (CR).